Programming Fundamentals

LAB 9 Spring 2025 SECTION: B8
Basic Pointers

Instructions:

- You must submit CPP files of the program in a folder, named your Registration Number.
- You must upload your lab tasks on CMS.
- All program codes should be written in C/C++. Students should use Visual studio compiler for coding.
- Indent and comment your code.
- Use meaningful variable names
- Plan your code carefully on a piece of paper before you implement it.

CLO NO	CLO STATEMENT	Blooms Taxonomy Level	PLO
1	Create solutions for real-world issues while staying within restrictions and making the most use of available resources by utilizing sophisticated problem-solving methods.	P3	5

1. Write a C++ program where you have to declare an integer variable to store price of car. Declare a pointer to store the address of this price variable. Now, ask user to enter price of car. Read this price from user using pointer and Display the 1^{st} installment (1^{st} installment is 12% of total car price + 15000 fixed charges) which is to be paid to buy this car. You can only use pointer for reading, calculation and printing.

   <u>Sample Ouput</u>

   Enter price of car: ***795000***
   1^{st} installment is: ***1,10,400/-***

2.Write a C++ program to ask user to enter First value, Second value (both are of type float), and an operator (to perform an arithmatic operations on two values). Declare 3 pointers which are pointing to these 3 data items. Now, perform relvant arithmatic operation (Addition, Subtraction, Multiplication, or Division) using only pointers and display result which is stored in another pointer which is pointing to another variable result.

<u>Sample Ouput</u>

Enter 1^{st} value: ***50***
Enter 2^{nd} Value: ***10***
*Enter an operator (+, -, *, or /):* ***/***
Division is: 5

3.Write a C++ program to declare an integer array of 50 elements. Fill this array using random function with values in range -25 to +75. Now, print address(s) of only those indexes which are having a value which is a prime No.

<u>Sample Ouput</u>

0	*1*	*2*	*3*	*48*	*49*
74	*-12*	*17*		*-11*	*54*

Address(s) of indexes with prime no. as value:
17: 0x6ffe14
11: 0x6df110

4.Write a program in C++ to find the maximum number between three numbers using a pointer.

Sample Ouput

Enter 1st value : ***54*** *Enter 2nd value:* ***17*** *Enter 3rd Value:* ***191*** *Largest value is:* ***191***

5.Write a C++ program to swap values of two variables using their pointers.

Note: *You can not use 3rd variable or 3rd pointer.*

Sample Ouput

Befor Swaping… *X contains: 25* *Y contains: 75* ***After Swaping…*** *X contains: 75* *Y contains: 25*

6.Write a C++ program that takes a 3-digit number from the user. The program should:

1. Check if the reverse of the number is equal to the original (palindrome check).
2. Print the square of the number if it is a palindrome. Otherwise, print the cube of the reversed number.

Function Requirements:

- `bool isPalindrome(int n)` *– checks if number is a palindrome.*
- `int reverseNumber(int n)` *– returns reversed number.*
- `void processNumber(int n)` *– handles logic and prints result.*
- `main()` *should only take input and call* `processNumber`*.*

Test Case 1:

Input:

Enter a 3-digit number: 121

Output:

The number is a palindrome.

Square: 14641

Test Case 2:

Input:

Enter a 3-digit number: 321

Output:

The number is not a palindrome.

Reversed number: 123

Cube: 1860867

7.Write a C++ program that:

1. Takes input for **number of guests**.
2. For each guest:
   - Takes input for **reversed name** and **days**.
   - Converts the reversed name into the **correct name**.
3. Finds the guest who will stay for the **most days**.
4. Displays the VIP guest's **name** and **number of days**.

Input Example:

Enter number of guests: 3

Enter reversed name and days: ecilA 3

Enter reversed name and days: boB 5

Enter reversed name and days: nhoJ 2

Output Example:

Guest: Bob

Days: 5

Question :You are tasked with sorting a set of numbers and analyzing the largest and smallest numbers in the list.

*Use **minimum 4 functions**:*

Your Tasks:

1. **Input the number of elements** the user will provide.
2. **Input the numbers** and store them in an integer array.
3. **Sort** the numbers in ascending order (without using any built-in sorting functions).
4. Find and display:
   - The **smallest number**.
   - The **largest number**.
   - The **sum** of all the numbers.
   - The **average** of the numbers.

Sample Input:

Enter the number of values: 5
Enter number: 9
Enter number: 4
Enter number: 12
Enter number: 7
Enter number: 1

Sample Output:

Sorted Numbers: 1 4 7 9 12

Smallest Number: 1

Largest Number: 12

Sum of Numbers: 33

Average of Numbers: 6.6

End of LAB 9 😊